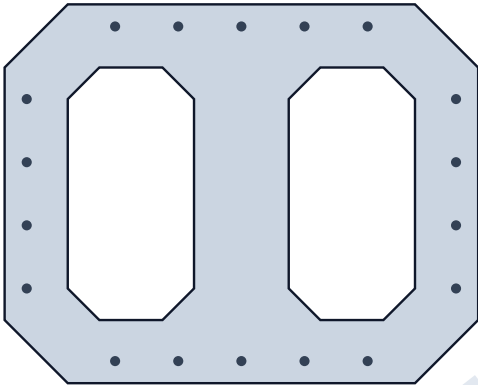


1. Input

Section, materials and resistance basis exactly as analysed.

Project information			
Project Name	PM-advanced (7) 2D reference case		
Client	Reference verification		
Company	P-M Column Designer		
Designed by	Engineering example	Checked by	Independent review
Address	Seoul, Korea	Date	2026-07-23
Calculation identity			
Section	PM-advanced (7) 2D — 1500x1200 .	Calculation profile	KDS 2024 — Stress-strain integratio.
Mechanics	Stress-strain integration	Resistance profile	KDS 2024 current set · resistance: K.
Profile edition	2024 partial-revision framework (effe.	Verification status	draft
Concrete model basis	Code-default	Method id	kds-2024-current-set-global-strength.
Units	mm · N · MPa; forces reported in kN,.	Sign convention	Compression positive; Mx = ΣF·(y-yc.
Section			
Cross-section (mm, about the net concrete centroid)		18 bars	Section properties
			Gross concrete area Ag1,120,000 mm²
			Reinforcement area As14,476 mm²
			Reinforcement ratio ρ1.293 %
			Bar count18
			Net centroid(0.0, 0.0) mm
			Bounding box1,500.0 × 1,200.0 mm
			Solids / holes1 / 2
			Perimeter9,262.7 mm
Materials			
Name	KDS C30 (workbook Input)	Standard	KDS
fck	30.0 MPa	Model	kds-parabolic
εcu	0.00330	εc0	0.00200
α	0.8500	Ec	28,417 MPa
Tension	ignored	Name	SD400 (workbook Input)
Standard	KDS	fy	400.0 MPa
Es	200,000 MPa	Model	elastic-perfectly-plastic
εy	0.002000	εu	not declared
Grades used	1		
Design resistance basis			
Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.700
φ tension	0.850	εt,limit	0.00500
Transition rule	fixed-or-yield-multiple	Axial cap	0.800 × factored compression pole
Sampling and numerical evidence			
Sampling mode	fixed	Active Design points	972
Active Design directions	36	Active Design stations	27
Active Nominal points	972	Active Nominal directions	36
Active Nominal stations	27	Direction tolerance	— %
Worst interpolation error	— %	Tolerance reached	yes
Concrete discretization	864 cells / 5196 points	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

Warning: Design profile status is draft; results are for review, not release.

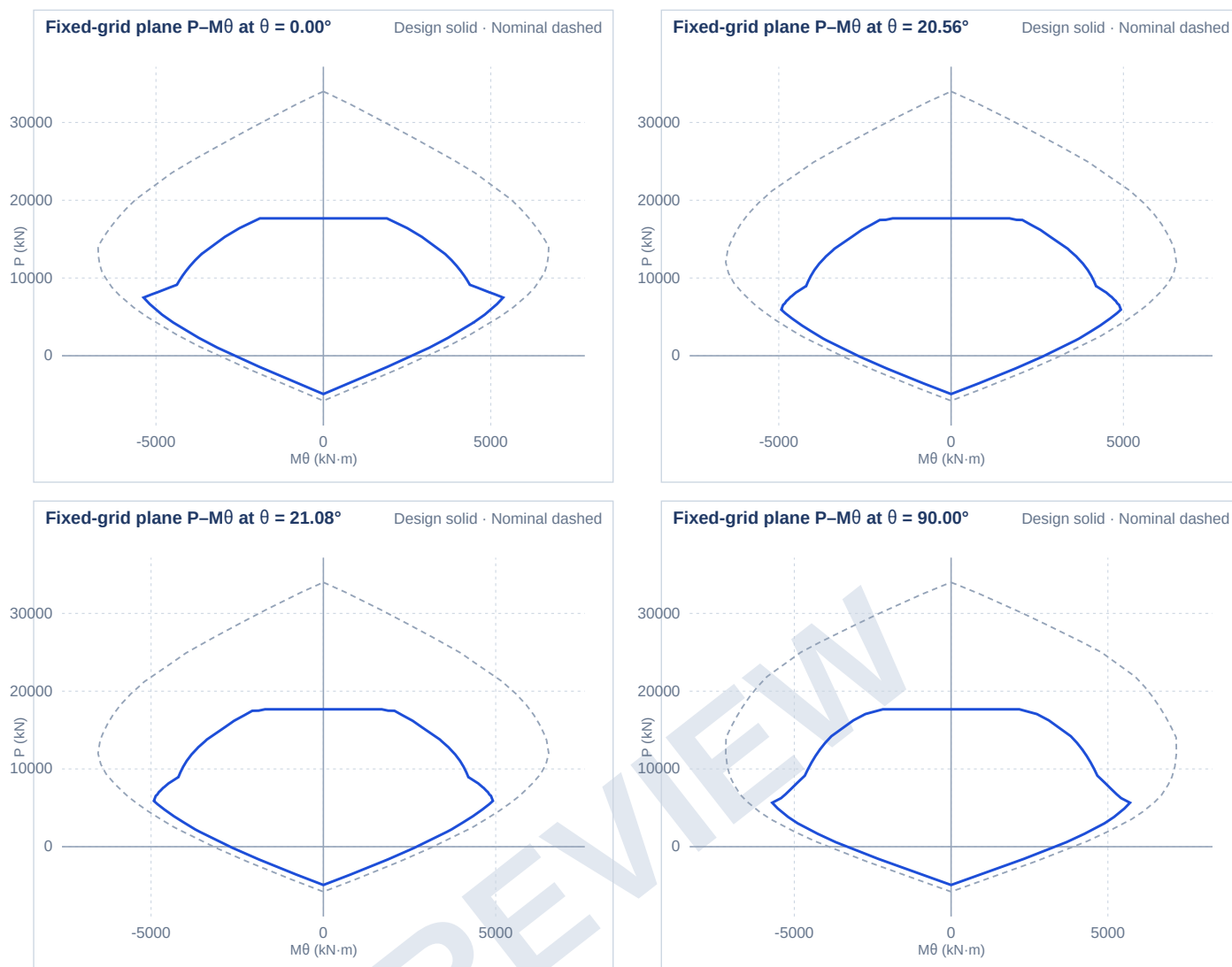
Reinforcement schedule

#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	32.0	-400.0	530.0	804.2	SD400 (workbook Input)
2	2	32.0	-200.0	530.0	804.2	SD400 (workbook Input)
3	3	32.0	0.0	530.0	804.2	SD400 (workbook Input)
4	4	32.0	200.0	530.0	804.2	SD400 (workbook Input)
5	5	32.0	400.0	530.0	804.2	SD400 (workbook Input)
6	6	32.0	-400.0	-530.0	804.2	SD400 (workbook Input)
7	7	32.0	-200.0	-530.0	804.2	SD400 (workbook Input)
8	8	32.0	0.0	-530.0	804.2	SD400 (workbook Input)
9	9	32.0	200.0	-530.0	804.2	SD400 (workbook Input)
10	10	32.0	400.0	-530.0	804.2	SD400 (workbook Input)
11	11	32.0	-680.0	300.0	804.2	SD400 (workbook Input)
12	12	32.0	-680.0	100.0	804.2	SD400 (workbook Input)
13	13	32.0	-680.0	-100.0	804.2	SD400 (workbook Input)
14	14	32.0	-680.0	-300.0	804.2	SD400 (workbook Input)
15	15	32.0	680.0	300.0	804.2	SD400 (workbook Input)
16	16	32.0	680.0	100.0	804.2	SD400 (workbook Input)
17	17	32.0	680.0	-100.0	804.2	SD400 (workbook Input)
18	18	32.0	680.0	-300.0	804.2	SD400 (workbook Input)

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 17670.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	φ	Class	UR	Verdict
1	ULS-1 workbook Newton d.	24942.9	3714.2	1431.8	21.1	0.650	compression-co.	1.539	INADEQUA.
2	ULS-2 same axial, half mo.	24942.9	1857.1	715.9	21.1	0.650	compression-co.	1.412	INADEQUA.
3	ULS-3 low axial, high mom.	5000.0	4000.0	1500.0	20.6	0.850	transition	0.877	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

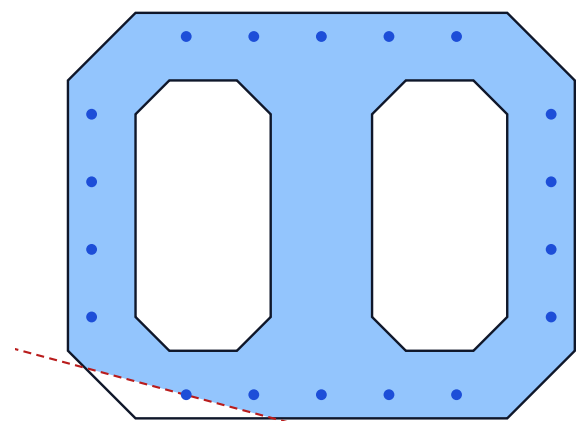
$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.5394 · fixed-P diagnostic —

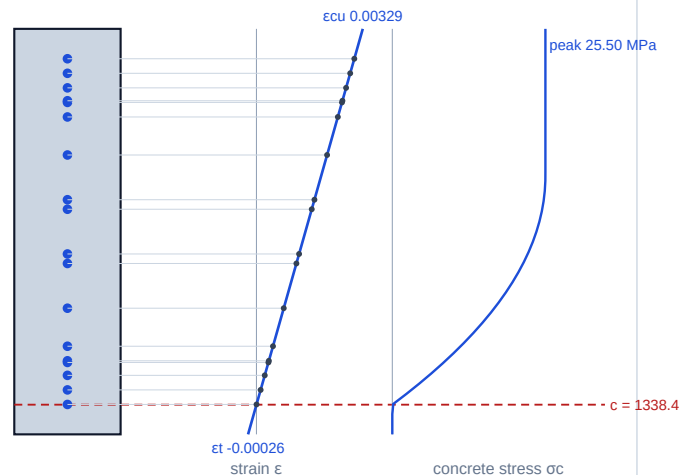
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta = 1444$

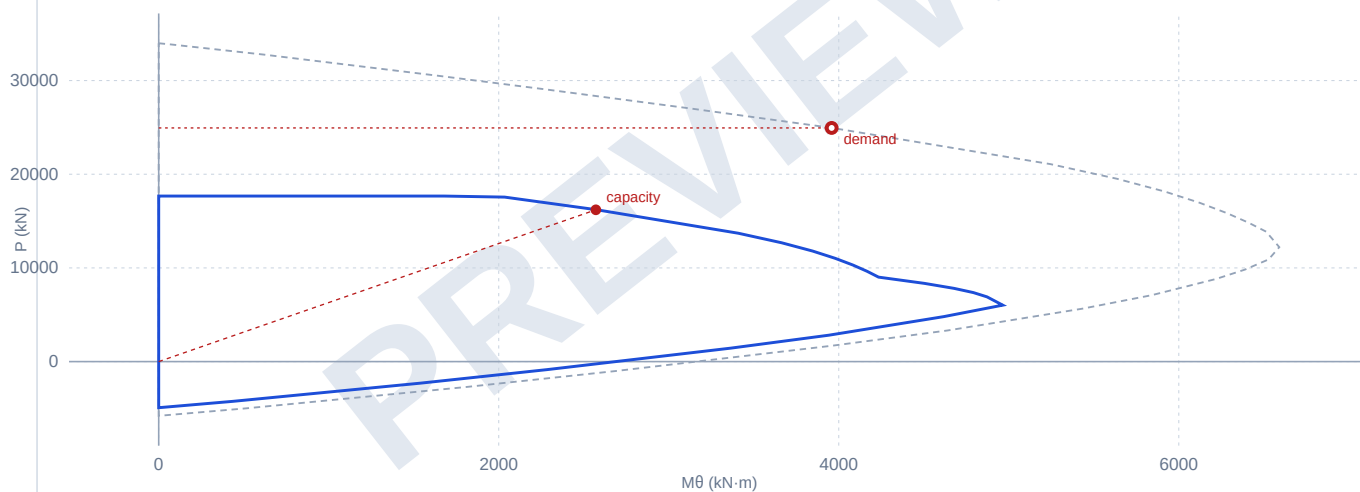


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P-Mβ at $\beta_{eq} = 15.019^\circ$, with the load point

solid = fixed Design · dashed = fixed Nominal · points are projected on β_{eq}



Concrete and resultants

Concrete force	21,577.4 kN	Concrete Mx	2,900.4 kN·m
Concrete My	1,122.3 kN·m	Steel force (gross)	3,629.5 kN
Displaced concrete	-264.0 kN	Steel net	3,365.5 kN
Response P	24,942.9 kN	Response Mx	3,714.2 kN·m
Response My	1,431.8 kN·m	Residual ΔP	1.01e-4 N
Residual ΔMx	3.15e-3 N·mm	Residual ΔMy	4.99e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.8	2.520e-3	400.00	321.70	—
2	-200.0	530.0	804.2	262.0	2.648e-3	400.00	321.70	—
3	0.0	530.0	804.2	210.1	2.775e-3	400.00	321.70	—
4	200.0	530.0	804.2	158.3	2.903e-3	400.00	321.70	—

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.5	3.030e-3	400.00	321.70	—
6	-400.0	-530.0	804.2	1337.6	1.971e-6	0.39	0.32	—
7	-200.0	-530.0	804.2	1285.8	1.295e-4	25.89	20.82	—
8	0.0	-530.0	804.2	1233.9	2.570e-4	51.39	41.33	—
9	200.0	-530.0	804.2	1182.1	3.845e-4	76.89	61.84	—
10	400.0	-530.0	804.2	1130.3	5.120e-4	102.39	82.35	—
11	-680.0	300.0	804.2	608.5	1.795e-3	359.09	288.80	—
12	-680.0	100.0	804.2	801.7	1.320e-3	264.06	212.37	—
13	-680.0	-100.0	804.2	994.8	8.451e-4	169.02	135.94	—
14	-680.0	-300.0	804.2	1188.0	3.699e-4	73.99	59.50	—
15	680.0	300.0	804.2	256.1	2.662e-3	400.00	321.70	—
16	680.0	100.0	804.2	449.2	2.187e-3	400.00	321.70	—
17	680.0	-100.0	804.2	642.4	1.712e-3	342.42	275.39	—
18	680.0	-300.0	804.2	835.6	1.237e-3	247.38	198.96	—

Solver and admissibility evidence

Strain plane ϵ_0	1.5162e-3	Curvature k_x	2.3759e-6
Curvature k_y	6.3748e-7	Neutral-axis line	164.9806°
Compression depth c	1,338.387 mm	Iterations	7
Relative residual	2.576e-12	Max concrete compression	0.003292
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	1.5394		

4.2 ULS-2 same axial, half moment (inside)

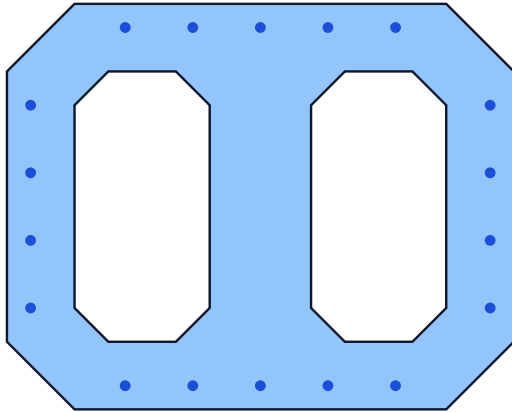
$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$ · $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

INADEQUATE · utilization 1.4116 · fixed-P diagnostic —

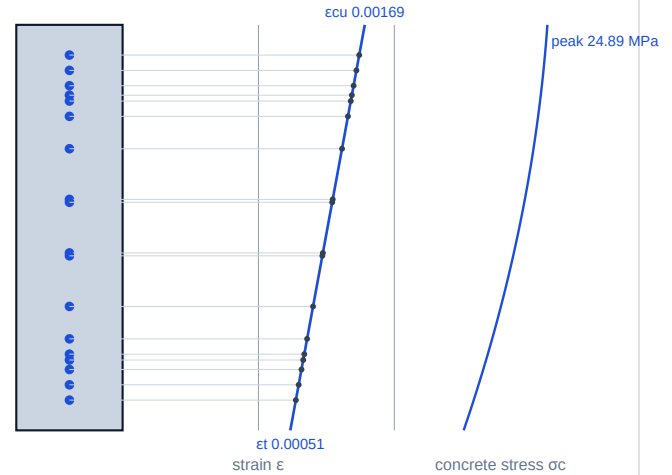
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta=1456$

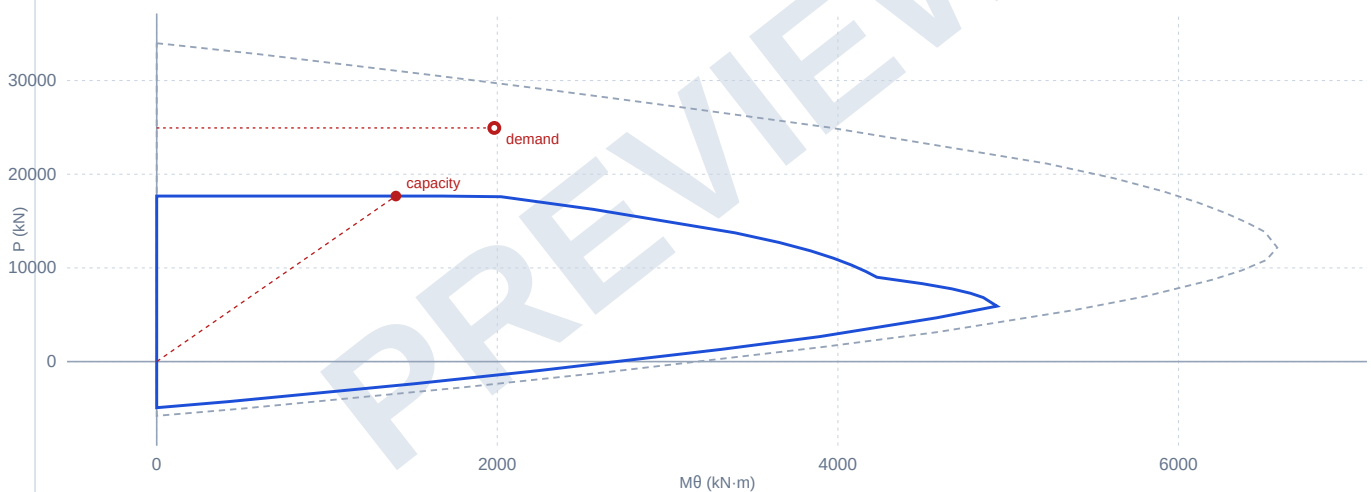


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct $P-M\beta$ at $\beta_{eq} = 15.968^\circ$, with the load point

solid = fixed Design · dashed = fixed Nominal · points are projected on β_{eq} $c = 2079.0$



Concrete and resultants

Concrete force	22,045.7 kN	Concrete Mx	1,476.8 kN·m
Concrete My	563.3 kN·m	Steel force (gross)	3,180.1 kN
Displaced concrete	-282.9 kN	Steel net	2,897.2 kN
Response P	24,942.9 kN	Response Mx	1,857.1 kN·m
Response My	715.9 kN·m	Residual $ \Delta P $	1.43e-5 N
Residual $ \Delta Mx $	5.19e-3 N·mm	Residual $ \Delta My $	1.92e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	328.6	1.423e-3	284.64	228.92	—
2	-200.0	530.0	804.2	273.6	1.468e-3	293.59	236.12	—
3	0.0	530.0	804.2	218.6	1.513e-3	302.54	243.32	—
4	200.0	530.0	804.2	163.6	1.557e-3	311.49	250.51	—

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	108.6	1.602e-3	320.43	257.71	—
6	-400.0	-530.0	804.2	1347.7	5.946e-4	118.92	95.64	—
7	-200.0	-530.0	804.2	1292.7	6.393e-4	127.87	102.84	—
8	0.0	-530.0	804.2	1237.7	6.841e-4	136.81	110.03	—
9	200.0	-530.0	804.2	1182.7	7.288e-4	145.76	117.23	—
10	400.0	-530.0	804.2	1127.7	7.735e-4	154.71	124.42	—
11	-680.0	300.0	804.2	626.8	1.181e-3	236.16	189.93	—
12	-680.0	100.0	804.2	819.1	1.024e-3	204.89	164.78	—
13	-680.0	-100.0	804.2	1011.4	8.681e-4	173.62	139.63	—
14	-680.0	-300.0	804.2	1203.7	7.118e-4	142.35	114.49	—
15	680.0	300.0	804.2	252.7	1.485e-3	297.00	238.86	—
16	680.0	100.0	804.2	444.9	1.329e-3	265.73	213.71	—
17	680.0	-100.0	804.2	637.2	1.172e-3	234.46	188.57	—
18	680.0	-300.0	804.2	829.5	1.016e-3	203.19	163.42	—

Solver and admissibility evidence

Strain plane ϵ_0	1.0984e-3	Curvature k_x	7.8173e-7
Curvature k_y	2.2369e-7	Neutral-axis line	164.0318°
Compression depth c	2,079.010 mm	Iterations	5
Relative residual	3.635e-13	Max concrete compression	0.001690
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	1.4116		

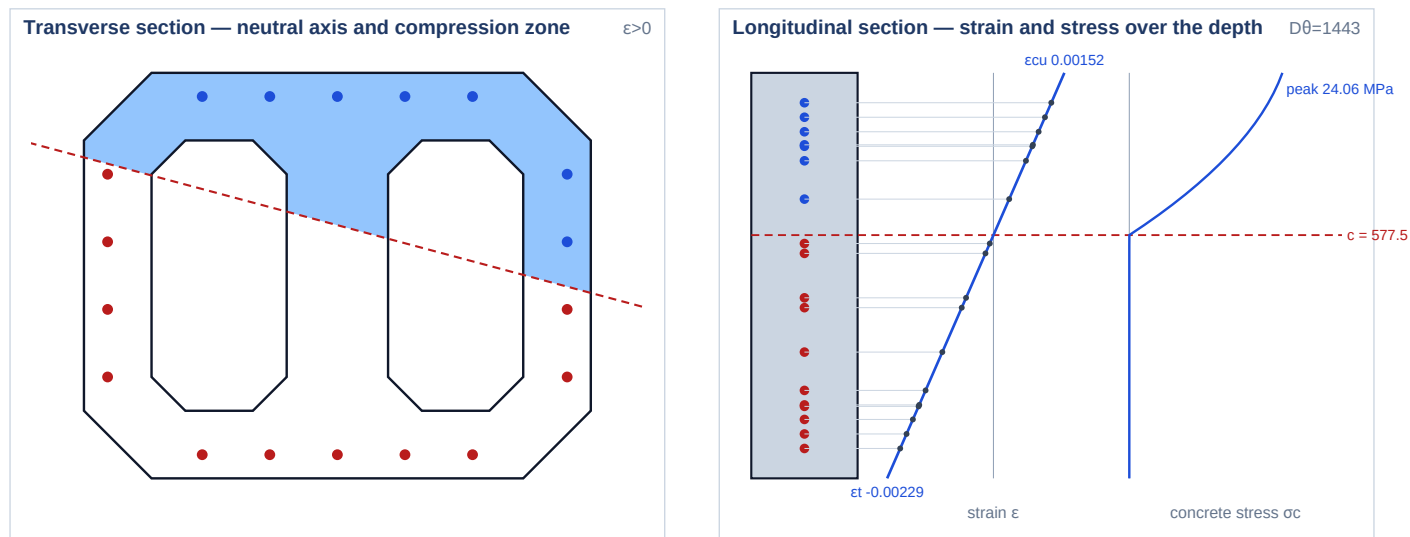
4.3 ULS-3 low axial, high moment

$P_u = 5000.0 \text{ kN}$ · $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 20.56^\circ$

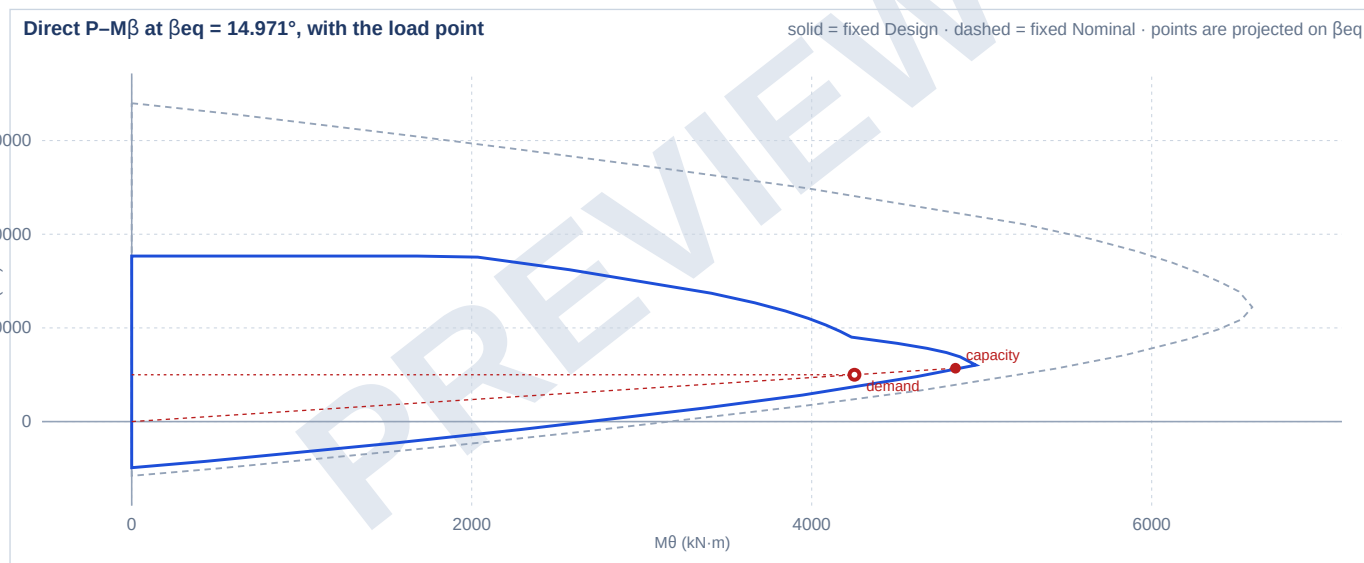
ADEQUATE · utilization **0.8774** · fixed-P diagnostic **0.9138**

Converged preview equilibrium inside the material domain.

Section state



Exact direct meridian at the equilibrium strain direction



Concrete and resultants

Concrete force	6,195.6 kN	Concrete Mx	2,728.2 kN·m
Concrete My	1,023.3 kN·m	Steel force (gross)	-1,101.1 kN
Displaced concrete	-94.5 kN	Steel net	-1,195.6 kN
Response P	5,000.0 kN	Response Mx	4,000.0 kN·m
Response My	1,500.0 kN·m	Residual ΔP	5.59e-9 N
Residual ΔMx	0.00e+0 N·mm	Residual ΔMy	9.54e-7 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.0	6.983e-4	139.66	112.33	—
2	-200.0	530.0	804.2	261.4	8.347e-4	166.94	134.26	—
3	0.0	530.0	804.2	209.7	9.711e-4	194.22	156.20	—
4	200.0	530.0	804.2	158.0	1.108e-3	221.50	178.14	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.4	1.244e-3	248.78	200.08	—
6	-400.0	-530.0	804.2	1337.1	-2.005e-3	-400.00	-321.70	—
7	-200.0	-530.0	804.2	1285.4	-1.869e-3	-373.74	-300.58	—
8	0.0	-530.0	804.2	1233.7	-1.732e-3	-346.46	-278.64	—
9	200.0	-530.0	804.2	1182.1	-1.596e-3	-319.18	-256.70	—
10	400.0	-530.0	804.2	1130.4	-1.460e-3	-291.91	-234.76	—
11	-680.0	300.0	804.2	607.6	-7.922e-5	-15.84	-12.74	—
12	-680.0	100.0	804.2	800.8	-5.893e-4	-117.86	-94.79	—
13	-680.0	-100.0	804.2	994.0	-1.099e-3	-219.88	-176.84	—
14	-680.0	-300.0	804.2	1187.2	-1.609e-3	-321.89	-258.88	—
15	680.0	300.0	804.2	256.2	8.483e-4	169.65	136.44	—
16	680.0	100.0	804.2	449.4	3.382e-4	67.64	54.40	—
17	680.0	-100.0	804.2	642.7	-1.719e-4	-34.38	-27.65	—
18	680.0	-300.0	804.2	835.9	-6.820e-4	-136.40	-109.70	—

Solver and admissibility evidence

Strain plane ϵ_0	-3.8060e-4	Curvature k_x	2.5504e-6
Curvature k_y	6.8197e-7	Neutral-axis line	165.0295°
Compression depth c	577.545 mm	Iterations	6
Relative residual	1.419e-16	Max concrete compression	0.001525
Concrete limit	0.003300	Max steel tension	0.002005
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	0.8774		